

Data Mining

TP: Data Cleaning and Exploratory Data Analysis (EDA)

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1. Context of the Work

Antibiotic resistance represents a major challenge in modern medicine, where bacteria adapt and become resistant to treatments that were once effective. The spread of multiresistant strains, defined as bacteria resistant to multiple antibiotic families, is particularly dangerous for hospitals and public health systems.

Antibiograms, which classify bacterial responses to antibiotics as Resistant (R), Sensitive (S), or Intermediate (I), provide the raw material for such studies. When combined with demographic, clinical, and contextual information, they allow researchers to identify critical resistance patterns, study associated risk factors, and design prediction models.

In this exercise, you will work with a dataset that simulates such data. However, before conducting any meaningful analysis, the dataset must undergo a full cleaning and preprocessing phase, since it contains multiple sources of noise, inconsistencies, and irrelevant variables.

2. Key Concepts

- **Antibiogram:** Laboratory test that measures the sensitivity of a bacterial strain to different antibiotics.
- **Antibiotic Families:** Groups of antibiotics with similar structures and mechanisms (e.g., β -lactams, Quinolones).
- **Resistance Values:** Standard categories are Resistant (R), Sensitive (S), and Intermediate (I), though noisy variants may exist.
- **Clinical Risk Factors:** Variables such as diabetes, hypertension, or hospitalization history that may influence resistance outcomes.
- **Multiresistance:** A strain is classified as multiresistant if it shows resistance in at least three antibiotic families.

3. Dataset Description

The dataset contains heterogeneous information combining biological results, demographic characteristics, and additional metadata.

- **Souches:** Identifier and species name of the bacterial strain. Contains inconsistencies and typographical errors.
- **Antibiotics:** One column per antibiotic (e.g., AMX/AMP, GEN, CIP). Values include R, S, I, or noisy symbols.
- **Age/Gender:** Combined information such as 45/F. Contains unrealistic ages, missing data, and inconsistent labels.
- **Risk Factors:** Diabetes, Hypertension, Hospitalization history, Infection frequency, etc., with inconsistent formats.
- **Collection_Date:** Laboratory test date, provided in multiple formats (ISO, European, textual) and sometimes missing.
- **Name:** Free-text patient names, inconsistent and non-standard. Irrelevant for analysis and raises privacy concerns.
- **Email:** Contact addresses, sometimes empty or malformed. No role in modeling.
- **Address:** Patient addresses in free-text form. Highly inconsistent, irrelevant for resistance analysis, and privacy-sensitive.
- **Notes:** Free-text comments by laboratory staff, with irregular formatting, symbols, and irrelevant information.
- **Collection_Date:** Although temporal data can be important, here the column is too noisy (mixed formats, missing values, future dates) and cannot be exploited without extensive normalization.

These variables must be identified and either removed or anonymized during preprocessing, as they can introduce bias, noise, or confidentiality issues.

4. Preprocessing Tasks

Cleaning and preparing the dataset involves several layers of transformation:

1. **Handling Bacterial Strains** Standardize species names by removing identifiers (e.g., S299) and correcting misspellings (e.g., E.coli → Escherichia coli).

2. **Antibiotic Values** Normalize values across all antibiotic columns: convert lowercase letters (r, s, i), harmonize labels (“Intermediate” → I), and replace invalid entries (“?”, “missing”).
3. **Age and Gender** Split the combined column into two. Detect and handle anomalies such as negative ages or unrealistic values. Standardize gender entries (M/F).
4. **Risk Factors** Recode categorical variables consistently: for instance, map {Yes, Y, True, 1} to “Yes” and {No, False, 0} to “No”. Replace ambiguous entries with missing values.
5. **Infection Frequency** Convert frequency labels to numerical categories (e.g., Never = 0, Rarely = 1, Regularly = 2, Often = 3). Handle anomalies (“error”, “?”, “missing”) as NaN.
6. **MultiResistance Indicator** Create a new binary variable: assign value = 1 if resistant to at least three antibiotic families, 0 otherwise.
7. **Irrelevant Columns** Drop metadata fields such as Name, Email, Address, Notes, and Collection_Date. These do not contribute to resistance analysis and may cause confidentiality issues.
8. **Duplicates and Consistency** Identify and remove duplicate rows. Verify column ordering to ensure a consistent schema for analysis.

5. Data Analysis Tasks

The cleaned dataset is now ready for exploratory data analysis (EDA). This section is divided into four major parts, each focusing on a different aspect of the dataset. For each part, you are asked to compute descriptive values, generate visualizations, and interpret the results.

5.1 Structure of the Dataset

The first step is to gain an overview of the dataset and its composition.

1. What is the total number of rows (patients/isolates) and columns (features) in the dataset?
2. Count how many columns correspond to antibiotics, demographic variables, clinical risk factors, and derived features.
3. How many unique bacterial species (**Souches**) are represented in the dataset?

4. What is the distribution of missing values per column? Which variables are most affected?
5. Are there duplicate rows in the dataset? If yes, how many, and how should they be handled?
6. Identify the most frequent bacterial species and the rarest species. How balanced is the dataset across species?
7. Check the value ranges of the numerical variables (**Age**, **Infection_Freq**). Are there anomalies or outliers?
8. Which categorical features contain the most unique values? Do you observe noisy categories?
9. Verify whether column order is consistent with the data schema. Does the structure make analysis easier or harder?
10. If the dataset were to be extended with more samples, which variables would likely cause scalability or standardization issues?

Suggested visualizations: bar charts for species counts, missing-value heatmap, pie chart for column type distribution.

5.2 Analysis of Features (independent exploration)

Here, the objective is to explore each feature independently, without yet considering resistance or multiresistance.

1. What is the distribution of **Age** across the dataset? Compute summary statistics (mean, median, min, max, standard deviation).
2. Plot the distribution of **AgeCategory** (Newborn, Child, Teenager, Young Adult, Senior Adult, Senior). Which group dominates?
3. What is the distribution of **Gender** (M/F)? Is the dataset balanced across genders?
4. For **Infection_Freq**, what is the frequency of each category (0,1,2,3)? How many missing values exist?
5. Count the number of patients with and without **Diabetes**. What proportion of the dataset does this represent?
6. Repeat the same for **Hypertension** and **Hospital_before**.
7. For bacterial species (**Souches**), what are the top 5 most common species, and how many samples do they represent?

8. Are there species that occur only once or twice (rare species)? How might this affect later analysis?
9. Compute the correlation (if meaningful) between numeric variables such as **Age** and **Infection_Freq**.
10. Which variables appear most evenly distributed, and which appear highly skewed? What might this imply for further modeling?

Suggested visualizations: histograms, boxplots for Age, bar charts for categorical features, pie charts for proportions.

5.3 Analysis of Features and Resistance

This section explores the relationships between patient characteristics, bacterial strains, and antibiotic resistance profiles.

1. For each antibiotic column, compute the counts of R, S, and I values. Which antibiotic shows the highest resistance rate?
2. Calculate the resistance percentage for each antibiotic family (β -lactams, Quinolones, Aminoglycosides, etc.). Which family is most affected?
3. Compare resistance distribution between male and female patients. Are there notable differences?
4. Compare resistance rates across **AgeCategory**. Which age group is most associated with resistant strains?
5. Does the presence of **Diabetes** influence resistance rates for any antibiotics? Provide evidence.
6. Repeat the same analysis for **Hypertension** and **Hospital_before**.
7. Are certain bacterial species (**Souches**) systematically more resistant than others? Identify species-antibiotic pairs with very high resistance.
8. For each **Souches**, compute the average resistance across all antibiotics. Which species appear most problematic?
9. Compare resistance rates between patients with low infection frequency vs high frequency. Do frequent infections correlate with higher resistance?
10. Check whether resistance is uniformly distributed across all antibiotics or whether some antibiotics are almost always resistant or sensitive.

11. (Optional advanced) Cluster strains based on resistance profiles. Do clear groups of bacteria emerge?

Suggested visualizations: grouped bar charts, heatmaps (species vs antibiotics), stacked bar charts for gender/age comparisons, radar plots for families.

5.4 Analysis of Features and Multi-Resistance

Focus now on the derived feature **MultiResistance**, which indicates whether a bacterial strain is resistant to at least three antibiotic families.

1. What proportion of patients are classified as multiresistant (**MultiResistance** = 1) versus non-multiresistant (**MultiResistance** = 0)?
2. Compare the distribution of **MultiResistance** between males and females. Which gender shows higher prevalence?
3. Compare multiresistance rates across **AgeCategory** groups. Which age group is most at risk?
4. Compute the average **Age** in both groups (multiresistant vs non-multiresistant). Are multiresistant cases associated with older patients?
5. What is the distribution of **MultiResistance** across bacterial species (**Souches**)? Which species dominate?
6. Does the presence of **Diabetes** increase the likelihood of multiresistance? Provide counts and percentages.
7. Does **Hypertension** correlate with higher multiresistance? Compare proportions.
8. Investigate the impact of prior hospitalization (**Hospital_before**) on multiresistance. Are hospitalized patients more often multiresistant?
9. Analyze the relationship between **Infection_Freq** and multiresistance. Do patients with more frequent infections have higher odds of multiresistance?
10. For each antibiotic family, calculate the proportion of resistant patients who are also multiresistant. Which families drive multiresistance the most?
11. Compare resistance profiles (per antibiotic) between multiresistant and non-multiresistant groups. Do certain antibiotics systematically separate the two groups?
12. Does the number of antibiotics tested for a patient influence the likelihood of being classified as multiresistant?

13. Is multiresistance concentrated in a few bacterial species, or is it more widely distributed across species?
14. Explore whether there is an interaction effect between **AgeCategory**, **Gender**, and multiresistance. For example, are older males more prone to multiresistance?
15. (Optional advanced) Propose a visualization combining multiple variables (e.g., **Souches**, **AgeCategory**, **Gender**) to study multiresistance distribution in a multi-dimensional way (heatmap, mosaic plot, 3D bar plot).

Suggested visualizations: pie chart for overall prevalence, stacked bar charts for demographic comparisons, bar chart of species vs multiresistance, grouped bars for comorbidities, heatmaps for multidimensional analysis.

6. Expected Outcomes

By the end of this assignment, you should be able to:

- Apply advanced data cleaning techniques on heterogeneous, noisy biomedical data.
- Create new, meaningful features such as categorical age groups and a multi-resistance indicator.
- Conduct a structured exploratory analysis to summarize and visualize resistance patterns.
- Develop critical thinking by interpreting resistance dynamics, multiresistance challenges, and associated clinical risk factors.